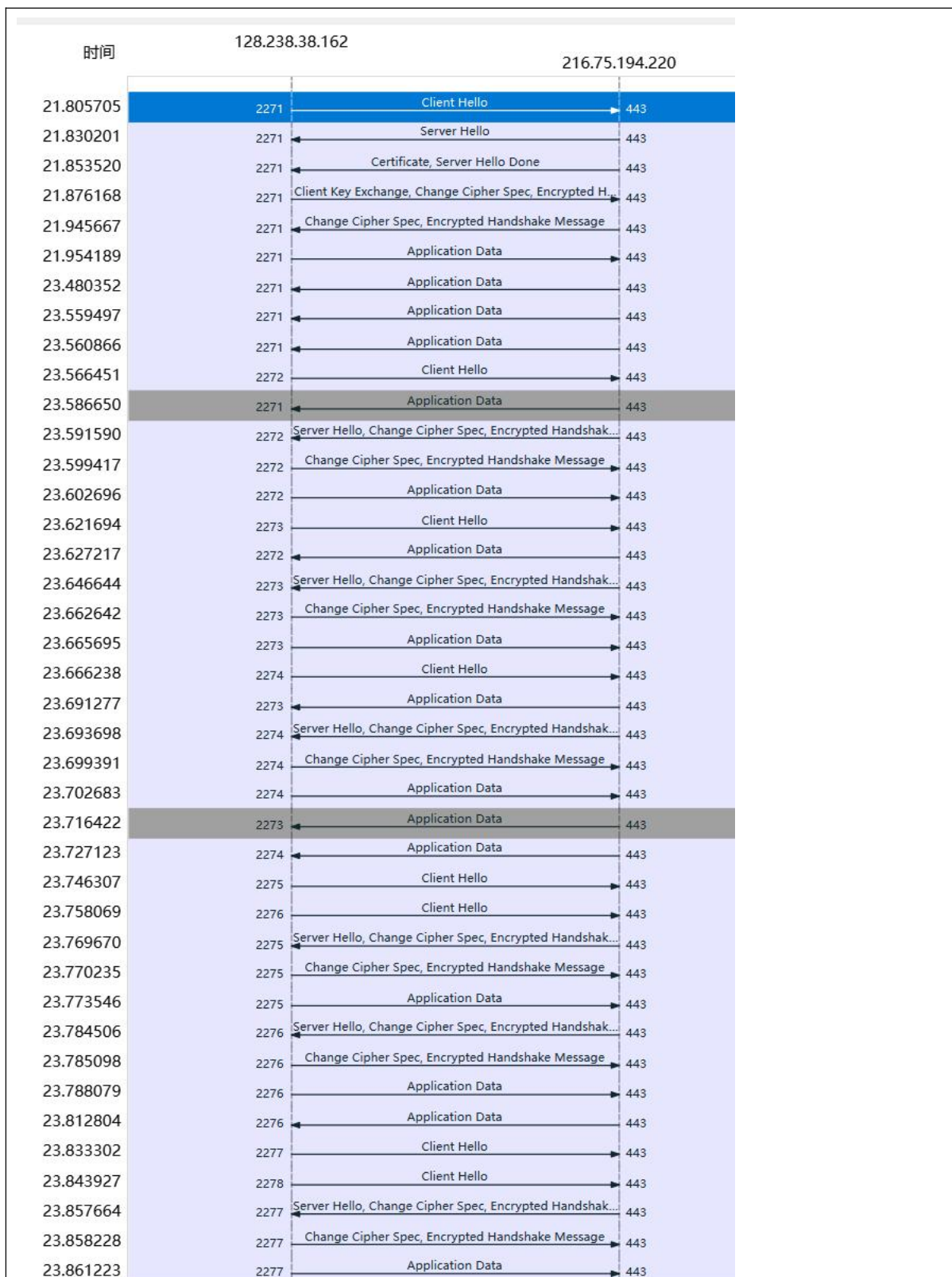


山东大学 计算机科学与技术 学院

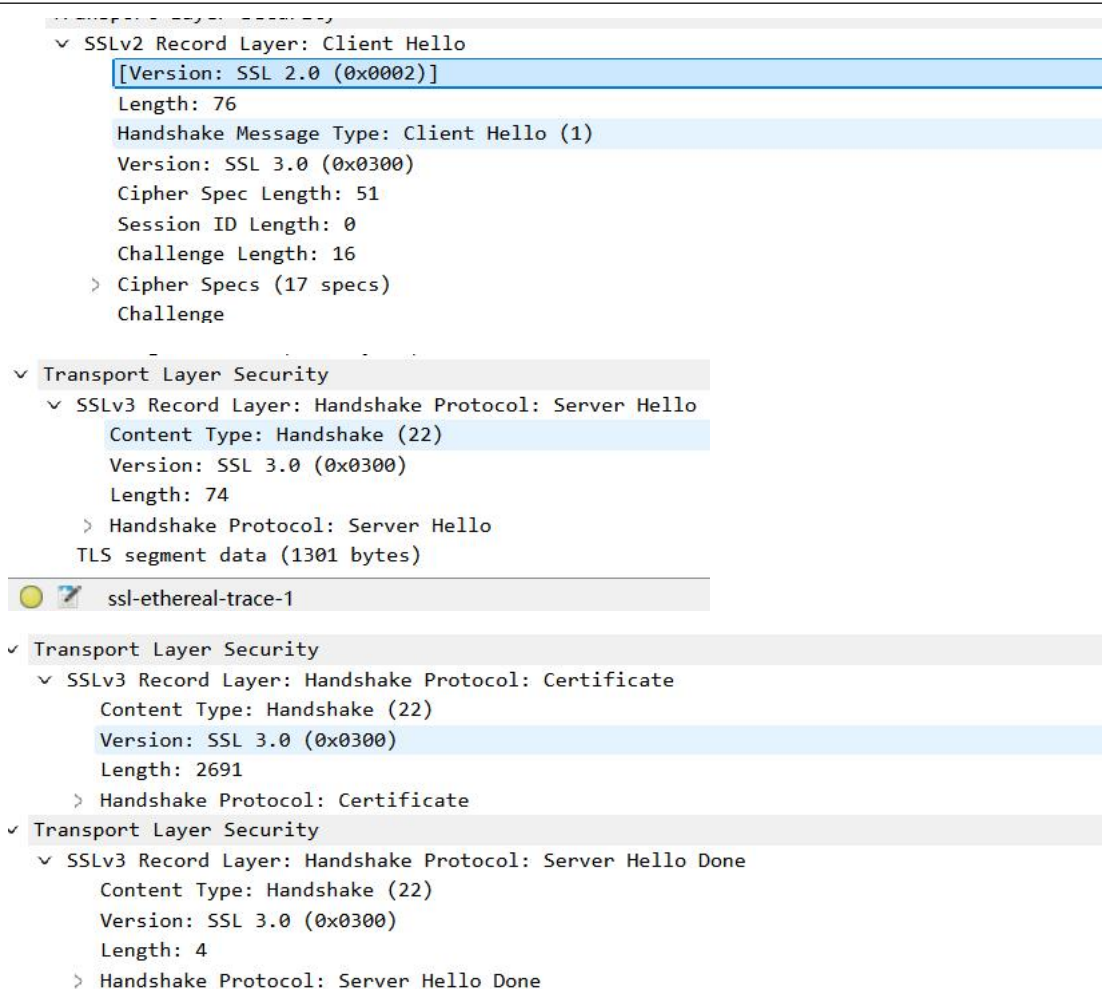
新兴网络技术与实践 课程实验报告

学号：202300130183	姓名：宋浩宇	班级：23 级智能班																																																															
实验题目：Wireshark Lab: SSL v8.0																																																																	
实验学时：2	实验日期：2025/6/11																																																																
实验目的：学习了解 SSL																																																																	
实验结果： 1. For each of the first 8 Ethernet frames, specify the source of the frame (client or server), determine the number of SSL records that are included in the frame, and list the SSL record types that are included in the frame. Draw a timing diagram between client and server, with one arrow for each SSL record. <table><thead><tr><th>No.</th><th>Time</th><th>Source</th><th>Destination</th><th>Protocol</th><th>Length</th><th>Info</th></tr></thead><tbody><tr><td>106</td><td>21.805705</td><td>128.238.38.162</td><td>216.75.194.220</td><td>SSLv2</td><td>132</td><td>Client Hello</td></tr><tr><td>108</td><td>21.830201</td><td>216.75.194.220</td><td>128.238.38.162</td><td>SSLv3</td><td>1434</td><td>Server Hello</td></tr><tr><td>111</td><td>21.853520</td><td>216.75.194.220</td><td>128.238.38.162</td><td>SSLv3</td><td>790</td><td>Certificate, Server Hello Done</td></tr><tr><td>112</td><td>21.876168</td><td>128.238.38.162</td><td>216.75.194.220</td><td>SSLv3</td><td>258</td><td>Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message</td></tr><tr><td>113</td><td>21.945667</td><td>216.75.194.220</td><td>128.238.38.162</td><td>SSLv3</td><td>121</td><td>Change Cipher Spec, Encrypted Handshake Message</td></tr><tr><td>114</td><td>21.954189</td><td>128.238.38.162</td><td>216.75.194.220</td><td>SSLv3</td><td>806</td><td>Application Data</td></tr><tr><td>122</td><td>23.480352</td><td>216.75.194.220</td><td>128.238.38.162</td><td>SSLv3</td><td>272</td><td>Application Data</td></tr><tr><td>149</td><td>23.559497</td><td>216.75.194.220</td><td>128.238.38.162</td><td>SSLv3</td><td>1367</td><td>Application Data</td></tr></tbody></table> 如图，前八个分别是客户端、服务端、服务端、客户端、服务端、客户端、服务端、服务端 有 Client Hello、Server Hello、Certificate、Client Key Exchange、Change Cipher Spec、Hypertext Transfer Protocol 这几种。			No.	Time	Source	Destination	Protocol	Length	Info	106	21.805705	128.238.38.162	216.75.194.220	SSLv2	132	Client Hello	108	21.830201	216.75.194.220	128.238.38.162	SSLv3	1434	Server Hello	111	21.853520	216.75.194.220	128.238.38.162	SSLv3	790	Certificate, Server Hello Done	112	21.876168	128.238.38.162	216.75.194.220	SSLv3	258	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message	113	21.945667	216.75.194.220	128.238.38.162	SSLv3	121	Change Cipher Spec, Encrypted Handshake Message	114	21.954189	128.238.38.162	216.75.194.220	SSLv3	806	Application Data	122	23.480352	216.75.194.220	128.238.38.162	SSLv3	272	Application Data	149	23.559497	216.75.194.220	128.238.38.162	SSLv3	1367	Application Data
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如图，这是 SSL 记录统计图

2. Each of the SSL records begins with the same three fields (with possibly different values). One of these fields is “content type” and has length of one byte. List all three fields and their lengths.



如图，content type、version、length 字段均已列出

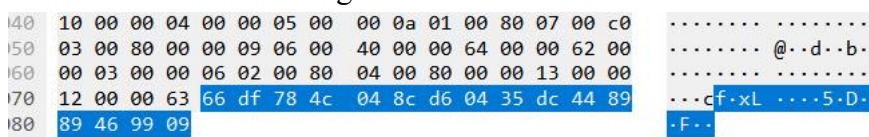
ClientHello Record:

3. Expand the ClientHello record. (If your trace contains multiple ClientHello records, expand the frame that contains the first one.) What is the value of the content type?



Content type 的值是 22

4. Does the ClientHello record contain a nonce (also known as a “challenge”)? If so, what is the value of the challenge in hexadecimal notation?



有这个字段，其十六进制值是 66 df 78 4c 04 8c d6 04 35 dc 44 89 89 46 99 09

5. Does the ClientHello record advertise the cyber suites it supports? If so, in the first listed suite, what are the public-key algorithm, the symmetric-key algorithm, and the hash algorithm?

```
✓ Cipher Specs (17 specs)
  Cipher Spec: TLS_RSA_WITH_RC4_128_MD5 (0x000004)
  Cipher Spec: TLS_RSA_WITH_RC4_128_SHA (0x000005)
  Cipher Spec: TLS_RSA_WITH_3DES_EDE_CBC_SHA (0x00000a)
  Cipher Spec: SSL2_RC4_128_WITH_MD5 (0x010080)
  Cipher Spec: SSL2_DES_192_EDE3_CBC_WITH_MD5 (0x0700c0)
  Cipher Spec: SSL2_RC2_128_CBC_WITH_MD5 (0x030080)
  Cipher Spec: TLS_RSA_WITH_DES_CBC_SHA (0x000009)
  Cipher Spec: SSL2_DES_64_CBC_WITH_MD5 (0x060040)
  Cipher Spec: TLS_RSA_EXPORT1024_WITH_RC4_56_SHA (0x000064)
  Cipher Spec: TLS_RSA_EXPORT1024_WITH_DES_CBC_SHA (0x000062)
  Cipher Spec: TLS_RSA_EXPORT_WITH_RC4_40_MD5 (0x000003)
  Cipher Spec: TLS_RSA_EXPORT_WITH_RC2_CBC_40_MD5 (0x000006)
  Cipher Spec: SSL2_RC4_128_EXPORT40_WITH_MD5 (0x020080)
  Cipher Spec: SSL2_RC2_128_CBC_EXPORT40_WITH_MD5 (0x040080)
  Cipher Spec: TLS_DHE_DSS_WITH_3DES_EDE_CBC_SHA (0x000013)
  Cipher Spec: TLS_DHE_DSS_WITH_DES_CBC_SHA (0x000012)
  Cipher Spec: TLS_DHE_DSS_EXPORT1024_WITH_DES_CBC_SHA (0x000063)
```

如图

TLS_RSA 表示密钥交换算法使用 RSA。

RC4_128 是对称加密算法，密钥长度为 128 位。

MD5 是哈希算法，用于消息认证。

ServerHello Record:

6. Locate the ServerHello SSL record. Does this record specify a chosen cipher suite? What are the algorithms in the chosen cipher suite?

```
✓ Transport Layer Security
  ✓ SSLv3 Record Layer: Handshake Protocol: Server Hello
    Content Type: Handshake (22)
    Version: SSL 3.0 (0x0300)
    Length: 74
  ✓ Handshake Protocol: Server Hello
    Handshake Type: Server Hello (2)
    Length: 70
    Version: SSL 3.0 (0x0300)
    > Random: 000000042dbed248b8831d04cc98c26e5badc4e267c391944f0f070ece57745
    Session ID Length: 32
    Session ID: 1bad05faba02ea92c64c54be4547c32f3e3ca63d3a0c86ddad694b45682da22f
    Cipher Suite: TLS_RSA_WITH_RC4_128_MD5 (0x0004)
    Compression Method: null (0)
    [JA3S Fullstring: 768,4,]
    [JA3S: 1f8f5a3d2fd435e36084db890693eafd]
  TLS segment data (1301 bytes)
```

指定了，和客户端的第一条是一样的 Cipher Spec: TLS_RSA_WITH_RC4_128_MD5 (0x000004)

TLS_RSA 表示密钥交换算法使用 RSA。

RC4_128 是对称加密算法，密钥长度为 128 位。

MD5 是哈希算法，用于消息认证。

7. Does this record include a nonce? If so, how long is it? What is the purpose of the client and server nonces in SSL?

```
✓ Random: 000000042dbed248b8831d04cc98c26e5badc4e267c391944f0f070ece57745
  GMT Unix Time: Jan 1, 1970 08:00:00.000000000 中国标准时间
  Random Bytes: 42dbed248b8831d04cc98c26e5badc4e267c391944f0f070ece57745
  Session ID Length: 32
```

有随机数，27 字节，目的是生成唯一的主密钥和会话密钥，防止重放攻击，确保通信的唯一性

和不可预测性，验证通信双方的实时性。

8. Does this record include a session ID? What is the purpose of the session ID?

Session ID Length: 32
Session ID: 1bad05faba02ea92c64c54be4547c32f3e3ca63d3a0c86ddad694b45682da22f

有会话 id，目的是用于会话恢复、会话管理、会话缓存、避免重复握手、避免重放攻击。

9. Does this record contain a certificate, or is the certificate included in a separate record. Does the certificate fit into a single Ethernet frame?

[JA3S Fullstring: 768,4,]
[JA3S: 1f8f5a3d2fd435e36084db890693eafd]

TLS segment data (1301 bytes)

有协议，协议内容直接包含在了记录里。

108	21.830201	216.75.194.220	128.238.38.162	SSLv3	1434 Server Hello
111	21.853520	216.75.194.220	128.238.38.162	SSLv3	790 Certificate, Server Hello Done

协议内容被切片了，用了两个以太网帧。

Client Key Exchange Record:

10. Locate the client key exchange record. Does this record contain a pre-master secret? What is this secret used for? Is the secret encrypted? If so, how? How long is the encrypted secret?

401 42.161162 216.75.194.220 128.238.38.162 SSLv3 412 Application Data

[Window size scaling factor: -2 (no window scaling used)]
Checksum: 0xc2d9 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
> [Timestamps]
> [SEQ/ACK analysis]
TCP payload (204 bytes)

▼ Transport Layer Security

▼ SSLv3 Record Layer: Handshake Protocol: Client Key Exchange

Content Type: Handshake (22)
Version: SSL 3.0 (0x0300)
Length: 132

▼ Handshake Protocol: Client Key Exchange

Handshake Type: Client Key Exchange (16)
Length: 128

▼ RSA Encrypted PreMaster Secret

Encrypted PreMaster [...]: bc49494729aa2590477fd059056ae78956c77b12af08b47c609e61f104b0fbf83e41c08dc910939cad1ece82e0dde250b99...

▼ SSLv3 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec

Content Type: Change Cipher Spec (20)
Version: SSL 3.0 (0x0300)
Length: 1
Change Cipher Spec Message

▼ SSLv3 Record Layer: Handshake Protocol: Encrypted Handshake Message

Content Type: Handshake (22)
Version: SSL 3.0 (0x0300)
Length: 56
Handshake Protocol: Encrypted Handshake Message

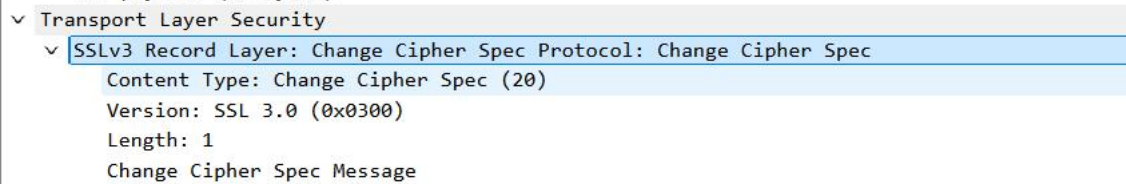
包含了，主要用于与客户端和服务器的 nonce 一起生成主密钥。主密钥随后被用于生成会话密钥，这些会话密钥将用于加密和认证通信数据。预主密钥在传输过程中是加密的。加密方式是 RSA 加密。长度是 128 字节。

Change Cipher Spec Record (sent by client) and Encrypted Handshake Record:

11. What is the purpose of the Change Cipher Spec record? How many bytes is the record in your trace?

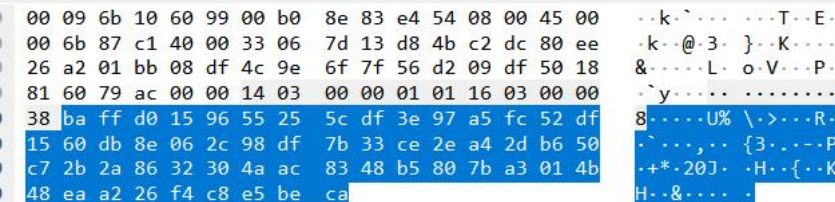
112	21.876168	128.238.38.162	216.75.194.220	SSLv3	258 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
113	21.945667	216.75.194.220	128.238.38.162	SSLv3	121 Change Cipher Spec, Encrypted Handshake Message
114	21.954189	128.238.38.162	216.75.194.220	SSLv3	806 Application Data

整个包的长度是 121 字节。



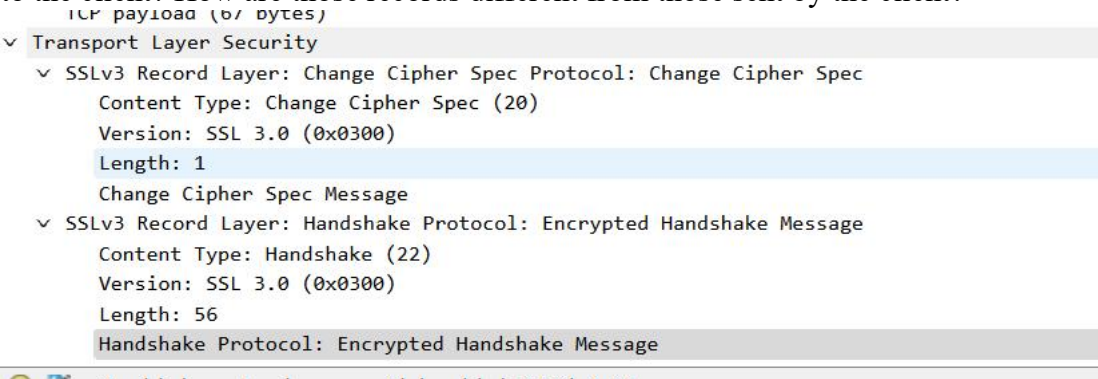
如果指的是具体的变更信息，那就是 6 字节

12. In the encrypted handshake record, what is being encrypted? How?



这是加密内容，通过 RSA 加密的。

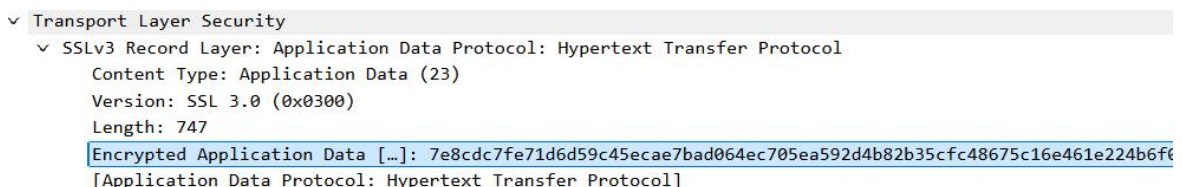
13. Does the server also send a change cipher record and an encrypted handshake record to the client? How are those records different from those sent by the client?



发送了，只看变更记录的话和客户端是一致的。

Application Data

14. How is the application data being encrypted? Do the records containing application data include a MAC? Does Wireshark distinguish between the encrypted application data and the MAC?



应用数据是通过 RSA 加密的，记录中不包含 MAC，wireshark 无法区分二者。

15. Comment on and explain anything else that you found interesting in the trace.

SSLv2	132	Client Hello
SSLv3	1434	Server Hello
SSLv3	790	Certificate, Server Hel
SSLv3	258	Client Key Exchange, Ch
SSLv3	121	Change Cipher Spec, Enc
SSLv3	806	Application Data
SSLv3	272	Application Data
SSLv3	1367	Application Data
SSLv3	1367	Application Data
SSLv3	156	Client Hello
SSLv3	1329	Application Data
SSLv3	200	Server Hello, Change Ci
SSLv3	121	Change Cipher Spec, Enc
SSLv3	470	Application Data
SSLv3	156	Client Hello
SSLv3	378	Application Data
SSLv3	200	Server Hello, Change Ci
SSLv3	121	Change Cipher Spec, Enc
SSLv3	476	Application Data
SSLv3	156	Client Hello
SSLv3	347	Application Data
SSLv3	200	Server Hello, Change Ci
SSLv3	121	Change Cipher Spec, Enc
SSLv3	472	Application Data
SSLv3	791	Application Data
SSLv3	445	Application Data

Client Hello 在最初发送时是 SSLv2，而在记录下服务端的 SSL 版本后，之后再建立链接使用的就是 SSLv3 了。

问题及收获：

SSL 是一种网络安全协议，主要用于在互联网通信中保障数据传输的安全性和隐私性。它的核心作用是通过加密技术对客户端和服务端之间传输的数据进行加密，防止信息在传输过程中被窃取或篡改，比如我们在网上购物输入信用卡信息、登录邮箱或社交账号时，SSL 协议就能保护我们的账号密码、个人隐私等重要信息不被坏人窃取。同时，它还能验证服务器的身份，确保我们访问的网站是真实的、可信的，而不是假冒的钓鱼网站，就像给网络通信建立了一个安全通道，让我们在浏览互联网时更安心。